



AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH
Faculty of Science and Technology
Department of Computer Science
CSC 1205 Object Oriented Programming - 1 (Section: All)

SET A

Mid Term Examination

Summer 2022-23

Total Marks: 100

Moderator: RIFATH MAHMUD

Time: 2 hours

General Instructions:

1. All the questions are based on OBE program outcomes. Question number 31 in Part B is mandatory and is mapped to CO number 1 (You must rewrite you Name, ID, Section, and Date on Part B).
2. Use pencil/pen to write the answer. Return the question paper, OMR sheet at the end of the examination.
3. Use ball point pen/Pencil to fill the OMR sheet. Do not staple/bend/fold the OMR sheet.

Specific Instructions:

1. Answer Part-A in the OMR. Answer Part-B, Part-C, Part-D & Part-E in the Question Paper within the given space.
2. Do not staple OMR with this question paper.
3. You may take signed loose sheet(s) from the Invigilators to answer Part-E. Staple only the loose sheet with this question paper that you have used for Part-E (code writing).

Name:		ID:
Department:	Section:	Proctor's Sign
Permit no:	Date:	

Part - A (Answer All)

MCQ

[1 x 30 = 30]

Use the following two options for questions 1 to 10.

- a) True b) False

1. Objects of a particular class maintain different memory locations for the instance variables. **T**
2. Encapsulation is achieved by declaring all the data members as public.
3. Procedural programming provides modularity and data security. **T**
4. In java, the main function is static because no objects are required to invoke it. **T**
5. Inheritance provides code reusability. **T**
6. In java, default access modifier can be applied at class and member level **T**
7. Static variables must be declared as public. **F**
8. More than one public class is allowed in a single java file. **F**
9. In java, objects of String class are mutable. **T //**
10. Multiple static blocks execute sequentially based on the declaration. **T**
11. A Java Byte Code can be run on any platform having a compatible version of _____.
a) JDK b) JRE c) JAVA Compiler d) Option a and b
12. Which one of the following is not a primitive data type?
a) int b) double c) String d) bool
13. In java, which option is valid for getting the size of a String?
a) length b) Size c) length() d) count
14. The String compareTo() method returns _____.
a) true, false b) -1, 0, 1 c) less than 0, 0, greater than 0 d) null
15. Which one will check whether str1 and str2 contain the same value or not?
a) str1.equal(str2) b) str1= str2 c) str1.equals(str2) d) str1== str2

16. The follow code segments outputs _____
 String s1 = "OOP1"; String s2 = "oop2";
 System.out.println(s1.compareTo(s2));
 a) 0 b) a value greater than 0 c) a value less than 0 d) false
17. What is the correct output of the following code segment?
 int i = (float)225;
 float f = i;
 System.out.println(f);
 a) 225 b) 225.0 c) 225.522 d) Compilation Error
18. Which of the following is an explicit call to the superclass's default constructor?
 a) default(); b) class(); c) super(); d) base();
19. _____ access modifier allows accessibility within all classes in the same package and within subclasses in other packages.
 a) private b) protected c) public d) package-private
20. Car is a type of Vehicle. Which OOP principle is indicated here?
 a) encapsulation b) inheritance c) polymorphism d) abstraction
21. Which variation of Inheritance is not supported by java.
 a) single b) multi-level c) hierarchical d) multiple
22. Identify the correct way of declaring constructor for a class named Entity
 a) Entity(){} b) public Entity(){} c) void Entity(){} d) Both a and b
23. Hide the last 5 digits of the phone number as +88017123*****
 Choose the appropriate parameters for the substring method to generate the output.
 String phoneNumber = "+8801712345678";
 System.out.println(phoneNumber.substring(____, ____)+ "*****");
 a) 1, 9 b) 0, 8 c) 0, 9 d) None
24. An array can store more than one type of data.
 a) Valid b) Invalid
25. Which of these is a legal declaration an array?
 a) int []arr = new int[]{-1, -2}; b) float []arr = (7, 8);
 c) char arr[] = [4, 5]; d) double arr[] = new double{2.5, 8.5};
26. Static block is usually used to initialize _____.
 a) Instance variables b) class variables c) objects d) default values
27. Which of these has the responsibility of providing controlled access to the data of a class?
 a) Parent class b) Attributes c) Methods d) this keyword
28. What is the correct Import statement to import Scanner class in java?
 a) import java.lang.Scanner; b) import java.util.Scanner; c) import util.Scanner d) None
29. Considering sc is an object of the Scanner class, which will be more appropriate for the following line?
 String message = sc._____;
 a) nextInt(); b) nextDouble() c) nextLine(); d) nextAnything();
30. "Officer is an Employee". Here ____
 a) Officer is the super class of Employee b) Employee is the base class of Officer
 c) Employee is the derived class of Officer d) Both Officer and Employee are child classes

OOP1MID Term Exam Summer 2023 **Part - B** **Question and Answer** **[1 x 10 = 10]**

Name:		Section:	
ID: 9		Date:	
Evaluation Criteria (To be filled up by Course Teacher)	Problem Analysis (5)	Proposed Solution (5)	Total (10)
Marks			

31. Problem Description: Consider the following class descriptions - Teacher and Student. Explain - problem(s) present in the context of Object Oriented Programming Principles. Provide a solution that incorporates appropriate OOP Principle(s).

Teacher
String name;
int age;
double salary;
void setName(String name)
void setAge(int age)
void setSalary(double salary)

Student
String name;
int age;
double CGPA;
void setName(String name)
void setAge(int age)
void setCGPA(double CGPA)

A. Identify Problem(s) in the given scenario in context of OOP principle(s). Write your answer below

B. Provide a justified solution that incorporates appropriate OOP principle(s). Write your answer below

```

8. public class Quiz{
    private static int x = 10;
    public int y;
    public Quiz(){
        System.out.println("Don't worry!");
        x--++x;
    }
    1) public Quiz(int x){
        this.y = y++;
        System.out.println("Programming is fun!");
        y--++y;
    }
    10 4- y += 2;}
    public void setX(int x){x = x;}
    public void setY(int y){this.y = y++;}
    public void display(){
        System.out.println("x: " + x + ", y: " + y);
    }
    public static void main(String args[]){
        1 Quiz obj = new Quiz();
        2 Quiz obj1 = new Quiz(7);
        String s1="don't loose hope!";
        String s2= new String("Keep");
        String s3 = new String("practicing ");
        String s4 = "Stay focused ";
        String s5 = new String();
        String s6 = "Stay";
        System.out.println(s5.isEmpty());
        System.out.println(s2.concat(" " + s3));
        System.out.println(s4);
        System.out.println(s1.substring(0,5) + s3);
        if(s2 == s6){
            obj1.setX(--Quiz.x);
            obj1.display();
        }
        else{
            obj1.setX(++Quiz.x);
            obj1.display();
        }
        if(s2.equals(s3)){
            obj1.setX(--Quiz.x);
            obj1.display();
        }
        else{
            obj1.setX(++Quiz.x);
            obj1.display();
        }
        s5 = s1.substring(6,16);
        System.out.println(s5);
        System.out.println((int)s3.charAt(0));
        if(s1.compareTo(s4) < 0){
            obj1.setY(--obj.y);
        }
        System.out.println("smaller");
        else{
            obj1.setY(obj.y--);
            System.out.println("greater or equal");
        }
        obj1.display();
    }
}

```

32. Complete the following code sample to ask the user what is the name and age of a person? Then store and display the details. (5)

```
import java.util.Scanner;
public class Person {
    private String name;
    private int age;
    public Person (String name, int age) {
        this.name = name;
        this.age = age;
    }
    public void displayInfo() {
        System.out.println("Name: " + name);
        System.out.println("Age: " + age);
    }
    public static void main(String[] args) {
        Scanner scanner = new Scanner (System.in);
        System.out.print("Enter name: ");
        String name = scanner.nextline();
        System.out.print("Enter age: ");
        int age = scanner.nextInt();
        Person person = new Person(name, age);
        person.displayInfo();
    }
}
```

33. Identify the type of type casting as implicit or explicit and provide examples for the following scenario. (5)

i. byte to short

Type of casting:

Example:

ii. float to double

Type of casting:

Example:

```

34. class ClassA{
    static{
        System.out.print("Loading Cl-
    }

    public ClassA(){
        System.out.println("Construct-
    }

    class ClassB{
        static{
            System.out.print("Loading Class: B ");
        }

        public ClassB(){
            System.out.println("
        }

        public static void main(String[] args){
            ClassB CB = new ClassB();
            CB = new ClassB();
            Class CA = new ClassA();

            static
            System.out.println("Loading Class: Main");
        }
    }

```

11. Outputs

```

35. public class Main {
    public static void main(String arg )
    int array[] = {10,27,2,15,7,9,30,21,12,21};
    float count = 0;
    float[] newarray = new float[10];

    for(int i=0;i<array.length;i++) {
        if(array[i]%3 == 0) {
            count++;
            newarray[i]=array[i];
        }

        System.out.println("Count:" + count);
        System.out.print("Elements:");

        for(int j = 0;j<newarray.length;j++) {
            System.out.println(newarray[j]);
        }
    }
}

```

21. Outputs

Part - D (Answer 2 out of 3)

Output Tracing

[2 x 15 = 30]

```
34. class ClassA{
    static{
        System.out.print("Loading Class: A ");
    }

    public ClassA(){
        System.out.println("Constructor: A");
    }
}

class ClassB{
    static{
        System.out.print("Loading Class: B ");
    }

    public ClassB(){
        System.out.println("Constructor: B");
    }
}

public class Main{
    public static void main(String [] args){
        ClassB cB = new ClassB();
        cB = new ClassB();
        ClassA cA = new ClassA();
    }
    static{
        System.out.println("Loading Class: Main");
    }
}
```

34. Outputs

Loading Class: A Loading Class: B
Loading Class: Main
Constructor: B
Constructor: A

```
35. public class Main {
    public static void main(String args[]){
        int array[] = {10,27,2,15,7,9,30,23,12,21};
        float count = 0;
        float[] newarray = new float[10];

        for(int i=0; i<array.length; i++) {
            if(array[i]%3 == 0) {
                count++;
                newarray[i]=array[i];
            }
        }
        System.out.println("Count: "+ count);
        System.out.print("Elements:");

        for(int j = 0; j<newarray.length; j++) {
            System.out.println(newarray[j]);
        }
    }
}
```

35. Outputs


```

36. import java.lang.*;
    public class Strings {
    public static void main(String args[] ) {
        String str1="OBJECT";
        String str2= new String("java.util.Scanner");
        String str3= str2.substring(4,10);
        String str4="OBJECT.oriented";
        String str5 = new String("PROGRAMMING");

        System.out.println("Length: "+str3.length());
        System.out.println(str1.compareTo(str4));
        System.out.println(str2.indexOf(','));
        System.out.print(str5.charAt(9));

        String str6=str1.toUpperCase(); OBJECT
        if(str1.equals(str6)){
            System.out.println("false"); false
        }
        else{
            System.out.println("true");
        }
    }
}

```

36. Outputs

Length: 6 ✓
 -9 ✓
 4 ✓
 N ✓
 15 ✓

Part - E

Code Writing

[1 x 25 = 25]

MobilePhone ✓	SmartPhone
String brand;	int cameraResolution;
String model;	int RAMSize;
double batteryCapacity;	int ROMSize;
MobilePhone()	String OSVersion;
MobilePhone(String brand,String model, double batteryCapacity)	SmartPhone() SmartPhone(String brand,String model, double batteryCapacity, int cameraResolution, int RAMSize, int ROMSize,String OSVersion)
void setBrand(String brand)	void setCameraResolution(int cameraResolution)
String getBrand()	int getCameraResolution()
void setModel(String model)	void setRAMSize(int RAMSize)
String getModel()	int getRAMSize()
void setBatteryCapacity(double batteryCapacity)	void setROMSize(int ROMSize)
double getBatteryCapacity()	int getROMSize()
void showDetails() -	void setOSVersion(String OSVersion)
	String getOSVersion() -
	void showDetails()

Note: You should complete the MobilePhone and SmartPhone class and after that write a Main Class and create two Sample objects of the SmartPhone class and demonstrate the use of different methods. (Start writing your code on loose sheet...)

Mr. Shairi Dhar	23-94871
Object Oriented Programming I (300)	F 2 A
12/02/2023	Room AL mod

Ref - E

```

public class MobilePhone {
    private String brand;
    private String model;
    private double batteryCapacity;

```

```

    MobilePhone() {

```

```

        MobilePhone(String brand, String model, double batteryCapacity) {

```

```

            this.brand = brand;

```

```

            this.model = model;

```

```

            this.batteryCapacity = batteryCapacity;

```

```

        public void setBrand(String brand) {

```

```

            this.brand = brand;

```

```

        public String getBrand() {

```

```

            return this.brand;

```

```

        public void setModel(String model) {

```

```

            this.model = model;

```

```

        public String getModel() {

```

```

            return this.model;

```